P05

Mechanical Impedance Methods and Mobility Analogies

Authors F. A. Firestone

Published Journal of Applied Physics (1938)

Paper Link https://doi.org/10.1063/1.1710430

Domain DOM_PHYS

Abstract

We present the mobility analogy mapping force to current and velocity to voltage. This topological correspondence allows mechanical networks to be solved with standard circuit loop and nodal theorems.

Abstract quoted from F. A. Firestone et al., Journal of Applied Physics (1938).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?